Milestone 4: API Specification

Stock News Trader

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1. Routes Overview

#	Method	Path	Description	Type
1	GET	`/api/companies`	List all companies with sector/industry	Simple
2	GET	`/api/companies/search`	Search companies by ticker or name	Simple
3	GET	`/api/companies/:ticker`	Company profile with latest price	Simple
4	GET	`/api/companies/:ticker/prices`	Historical OHLCV price data for charting	Simple
5	GET	`/api/stocks/top-gainers`	Top 10 gainers on a given date (CTE + window)	**Complex**
6	GET	`/api/sectors/:sector/companies`	Companies filtered by sector	Simple
7	GET	`/api/companies/:ticker/news`	Latest news articles for a company	Simple
8	GET	`/api/news/trending`	Most-mentioned companies in recent news	**Complex**
9	GET	`/api/sectors/returns`	Average return by sector on a given date (CTE + agg)	**Complex**
10	GET	`/api/sectors`	List all sectors	Simple
A1	POST	`/api/users`	Create a new user account	Auxiliary
A2	POST	`/api/users/login`	Authenticate a user	Auxiliary

2. Route Definitions

2.1. Route 1: List All Companies

`GET /api/companies`

Description: Listing all companies with their sector and industry names. Maps to Milestone 3 Query 1.

Request Parameters

None.

SQL Query

```sql
SELECT c.company_id, c.ticker, c.company_name,
       i.industry_name, s.sector_name
FROM company c
JOIN industry i ON c.industry_id = i.industry_id
JOIN sector s   ON i.sector_id   = s.sector_id
ORDER BY c.ticker
```

Response Schema

Field	Data Type	Description
`company_id`	integer	Internal company identifier
`ticker`	string	Stock ticker symbol
`company_name`	string	Full company name
`industry_name`	string	Industry the company belongs to
`sector_name`	string	Sector the company belongs to

HTTP Status Codes

Status	Description
`200 OK`	Returns array of all companies
`500 Internal Server Error`	Unexpected database error

2.2. Route 2: Search Companies

`GET /api/companies/search`

Description: Searching companies by ticker or company name using a case-insensitive prefix match. Intended for autocomplete. Maps to Milestone 3 Query 2.

Request Parameters

Name	Param Type	Data Type	Required?	Description
`q`	query	string	Required	Search prefix (e.g., `AAPL` or `Apple`)

SQL Query

```
SELECT company_id, ticker, company_name
FROM company
WHERE ticker ILIKE ($1 || '%')
   OR company_name ILIKE ($1 || '%')
ORDER BY ticker
```

Response Schema

Field	Data Type	Description
`company_id`	integer	Internal company identifier
`ticker`	string	Stock ticker symbol
`company_name`	string	Full company name

HTTP Status Codes

Status	Description
`200 OK`	Returns array of matched companies (may be empty)
`400 Bad Request`	`q` param is missing or empty
`500 Internal Server Error`	Unexpected database error

2.3. Route 3: Company Profile

`GET /api/companies/:ticker`

Description: Fetching a single company's profile including sector, industry, exchange, and CIK.

Request Parameters

Name	Param Type	Data Type	Required?	Description
`ticker`	path	string	Required	Stock ticker symbol (e.g., `AAPL`)

SQL Query

```sql
SELECT c.company_id, c.ticker, c.company_name,
       c.exchange, c.cik,
       i.industry_name, s.sector_name
FROM company c
JOIN industry i ON c.industry_id = i.industry_id
JOIN sector s   ON i.sector_id   = s.sector_id
WHERE c.ticker = $1
```

Response Schema

Field	Data Type	Description
`company_id`	integer	Internal company identifier
`ticker`	string	Stock ticker symbol
`company_name`	string	Full company name
`exchange`	string	Listing exchange
`cik`	string	SEC CIK identifier
`industry_name`	string	Industry name
`sector_name`	string	Sector name

HTTP Status Codes

Status	Description
`200 OK`	Company profile returned
`404 Not Found`	Ticker does not exist in the database
`500 Internal Server Error`	Unexpected database error

2.4. Route 4: Historical Prices

`GET /api/companies/:ticker/prices`

Description: Retrieving historical OHLCV price data for a company, ordered by date. Used for rendering time-series charts. Maps to Milestone 3 Query 4.

Request Parameters

Name	Param Type	Data Type	Required?	Description
`ticker`	path	string	Required	Stock ticker symbol

SQL Query

```
SELECT c.company_name, c.ticker,
       pd.trading_date, pd.open, pd.high, pd.low,
       pd.close, pd.adj_close, pd.volume
FROM price_daily pd
JOIN company c ON c.company_id = pd.company_id
WHERE c.ticker = $1
ORDER BY pd.trading_date
```

Response Schema

Field	Data Type	Description
`company_name`	string	Company name
`ticker`	string	Ticker symbol
`trading_date`	date	Trading date
`open`	float	Opening price
`high`	float	Day high
`low`	float	Day low
`close`	float	Closing price
`adj_close`	float	Adjusted close
`volume`	integer	Trading volume

HTTP Status Codes

Status	Description
`200 OK`	Returns array of daily price records
`404 Not Found`	Ticker does not exist
`500 Internal Server Error`	Unexpected database error

2.5. Route 5: Top 10 Gainers (Complex)

`GET /api/stocks/top-gainers`

Description: Returning the top 10 stocks by percentage gain on a given trading date compared to the previous day. Using CTEs and a multi-table join. Maps to Milestone 3 Query 3.

Request Parameters

Name	Param Type	Data Type	Required?	Description
`date`	query	date (YYYY-MM-DD)	Required	The trading date to evaluate

SQL Query

```sql
WITH latest AS (
    SELECT DISTINCT company_id, close, trading_date
    FROM price_daily
    WHERE trading_date = $1
),
prev AS (
    SELECT DISTINCT company_id, close AS prev_close
    FROM price_daily
    WHERE trading_date = $1::date - INTERVAL '1 day'
)
SELECT c.ticker, l.close, p.prev_close,
       ((l.close - p.prev_close) / p.prev_close) * 100
           AS pct_change
FROM latest l
JOIN prev p    ON l.company_id = p.company_id
JOIN company c ON c.company_id = l.company_id
ORDER BY pct_change DESC
LIMIT 10
```

Response Schema

Field	Data Type	Description
`ticker`	string	Stock ticker symbol
`close`	float	Closing price on the selected date
`prev_close`	float	Previous day's closing price
`pct_change`	float	Percentage change from previous close

HTTP Status Codes

Status	Description
`200 OK`	Returns top 10 gainers
`400 Bad Request`	`date` param is missing or not a valid date
`500 Internal Server Error`	Unexpected database error

2.6. Route 6: Companies by Sector

`GET /api/sectors/:sector/companies`

Description: Listing all companies belonging to a given sector, with their ticker and industry. Maps to Milestone 3 Query 5.

Request Parameters

Name	Param Type	Data Type	Required?	Description
`sector`	path	string	Required	Sector name (e.g., `Technology`)

SQL Query

```
SELECT c.company_id, c.ticker, c.company_name
FROM company c
JOIN industry i ON c.industry_id = i.industry_id
JOIN sector s   ON i.sector_id   = s.sector_id
WHERE s.sector_name = $1
ORDER BY c.ticker
```

Response Schema

Field	Data Type	Description
`company_id`	integer	Internal company identifier
`ticker`	string	Stock ticker symbol
`company_name`	string	Full company name

HTTP Status Codes

Status	Description
`200 OK`	Returns array of companies in the sector (may be empty)
`500 Internal Server Error`	Unexpected database error

2.7. Route 7: Company News

`GET /api/companies/:ticker/news`

Description: Retrieving the 10 most recent news articles mentioning a company. Joining `news_article` and `article_mention`. Maps to Milestone 3 Query 6.

Request Parameters

Name	Param Type	Data Type	Required?	Description
`ticker`	path	string	Required	Stock ticker symbol

SQL Query

```sql
SELECT c.company_name, c.ticker,
       na.title, na.source, na.published_at, na.url
FROM news_article na
JOIN article_mention am ON na.article_id = am.article_id
JOIN company c          ON c.company_id  = am.company_id
WHERE c.ticker = $1
ORDER BY na.published_at DESC
LIMIT 10
```

Response Schema

Field	Data Type	Description
`company_name`	string	Company name
`ticker`	string	Ticker symbol
`title`	string	Article headline
`source`	string	News source name
`published_at`	timestamptz	Publication timestamp
`url`	string	Article URL

HTTP Status Codes

Status	Description
`200 OK`	Returns array of news articles (may be empty)
`404 Not Found`	Ticker does not exist
`500 Internal Server Error`	Unexpected database error

2.8. Route 8: Trending Companies in News (Complex)

`GET /api/news/trending`

Description: Returning the top 10 most-mentioned companies in news articles over the past 30 days. Using aggregation with `GROUP BY` and a date-range filter. Maps to Milestone 3 Query 7.

Request Parameters

None.

SQL Query

```sql
SELECT c.ticker, COUNT(*) AS mention_count
FROM article_mention am
JOIN company c      ON am.company_id = c.company_id
JOIN news_article na ON na.article_id = am.article_id
WHERE na.published_at >= CURRENT_DATE - INTERVAL '30 days'
GROUP BY c.ticker
ORDER BY mention_count DESC
LIMIT 10
```

Response Schema

Field	Data Type	Description
`ticker`	string	Stock ticker symbol
`mention_count`	integer	Number of article mentions in the past 30 days

HTTP Status Codes

Status	Description
`200 OK`	Returns top 10 trending companies
`500 Internal Server Error`	Unexpected database error

2.9. Route 9: Sector Returns (Complex)

`GET /api/sectors/returns`

Description: Computing the average daily return by sector on a given date compared to the previous day. Using CTEs, multi-table joins, and aggregation across sector/industry/company/price tables. Maps to Milestone 3 Query 8.

Request Parameters

Name	Param Type	Data Type	Required?	Description
`date`	query	date (YYYY-MM-DD)	Required	The trading date to evaluate
`sector`	query	string	Optional	Filter to a specific sector

SQL Query

```sql
WITH latest AS (
    SELECT DISTINCT company_id, close
    FROM price_daily
    WHERE trading_date = $1
),
prev AS (
    SELECT DISTINCT company_id, close AS prev_close
    FROM price_daily
    WHERE trading_date = $1::date - INTERVAL '1 day'
)
SELECT s.sector_name,
       AVG((l.close - p.prev_close) / p.prev_close * 100)
           AS avg_return_pct
FROM latest l
JOIN prev p    ON l.company_id = p.company_id
JOIN company c ON c.company_id = l.company_id
JOIN industry i ON c.industry_id = i.industry_id
JOIN sector s   ON i.sector_id   = s.sector_id
WHERE ($2 IS NULL OR s.sector_name = $2)
GROUP BY s.sector_name
ORDER BY avg_return_pct DESC
```

Response Schema

Field	Data Type	Description
`sector_name`	string	Sector name
`avg_return_pct`	float	Average return percentage across companies in the sector

HTTP Status Codes

Status	Description
`200 OK`	Returns sector return data
`400 Bad Request`	`date` param is missing or not a valid date
`500 Internal Server Error`	Unexpected database error

2.10. Route 10: List Sectors

`GET /api/sectors`

Description: Returning all sectors. Used for populating filter dropdowns in the frontend.

Request Parameters

None.

SQL Query

```
SELECT sector_id, sector_name
FROM sector
ORDER BY sector_name
```

Response Schema

Field	Data Type	Description
`sector_id`	integer	Internal sector identifier
`sector_name`	string	Sector name

HTTP Status Codes

Status	Description
`200 OK`	Returns array of all sectors
`500 Internal Server Error`	Unexpected database error

3. Auxiliary Routes

3.1. Route A1: Create User

`POST /api/users`

Description: Creating a new user account. Supporting future watchlist and preference features.

Request Parameters

Name	Param Type	Data Type	Required?	Description
`username`	body	string	Required	Desired username
`email`	body	string	Required	User email address
`password`	body	string	Required	Password (hashed server-side)

SQL Query

```
INSERT INTO users (username, email, password_hash)
VALUES ($1, $2, $3)
RETURNING user_id, username
```

Response Schema

Field	Data Type	Description
`user_id`	integer	Newly created user's ID
`username`	string	Confirmed username

HTTP Status Codes

Status	Description
`201 Created`	User account created
`400 Bad Request`	Required field missing
`409 Conflict`	Username or email already exists
`500 Internal Server Error`	Unexpected database error

3.2. Route A2: User Login

`POST /api/users/login`

Description: Authenticating an existing user by username and password.

Request Parameters

Name	Param Type	Data Type	Required?	Description
`username`	body	string	Required	Existing username
`password`	body	string	Required	Password to verify

SQL Query

```
SELECT user_id, password_hash
FROM users
WHERE username = $1
```

Response Schema

Field	Data Type	Description
`token`	string	Session token
`user_id`	integer	Authenticated user's ID

HTTP Status Codes

Status	Description
`200 OK`	Login successful
`400 Bad Request`	Required field missing
`401 Unauthorized`	Incorrect password
`404 Not Found`	Username does not exist
`500 Internal Server Error`	Unexpected database error